# *SWAMP*: A Subsurface Wireless Agricultural Monitoring Platform


## Abstract

We present *SWAMP*, a low-cost wireless platform for non-invasive subsurface agricultural sensing using commodity radio-frequency measurements. *SWAMP* combines repeatable wideband channel frequency response (CFR) sweeps for growth monitoring with standardized Long Term Evolution (LTE) link-quality indicators for underground localization and auxiliary condition sensing. A lightweight Transmission Control Protocol (TCP) coordination protocol enables repeatable stepped-frequency CFR acquisition without Global Positioning System-disciplined oscillators. On top of this measurement pipeline, *SWAMP* extracts four interpretable CFR features for growth-stage inference and fuses five LTE indicators through constrained optimization to recover tuber probability maps. We evaluate *SWAMP* in a 45-day greenhouse sweet potato study spanning two soil recipes and two moisture regimes, together with auxiliary experiments on band selection, tuber-count discrimination, and moisture/fertility sensitivity. *SWAMP* achieves 83.0–91.0% accuracy for three-stage growth monitoring and 87.5–95.0% accuracy for 5 cm-grid underground localization, while requiring only 10–20 minutes of setup and roughly $3,000–$4,000 in hardware. Our results show that commodity wireless links can support practical non-invasive subsurface crop sensing in controlled but soil-diverse environments.



## Keywords

Commodity sensing, subsurface sensing, agriculture, LTE metrics, software-defined radio, channel frequency response, underground localization, soil condition sensing

**ACM Reference Format:**

. 2026. *SWAMP*: A Subsurface Wireless Agricultural Monitoring Platform. In . ACM, New York, NY, USA, 14 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn


## 1 Introduction

A substantial fraction of economically important crop biomass is formed belowground. For root and tuber crops such as potato and sweet potato, the yield-forming organs remain physically inaccessible throughout much of the growth cycle. As a result, crop management and breeding studies often rely on aboveground proxies, sparse manual inspection, or destructive sampling at a few discrete time points [19, 30, 38]. This limits the ability to observe tuber initiation, bulking, and spatial distribution during growth, even though these hidden states directly affect irrigation timing, fertilization, harvest planning, and breeding selection. Specialized root-phenotyping platforms and newer non-destructive underground monitoring methods have begun to address this gap, but many remain difficult to deploy in routine greenhouse workflows or require dedicated sensing infrastructure [28, 34, 41, 44].

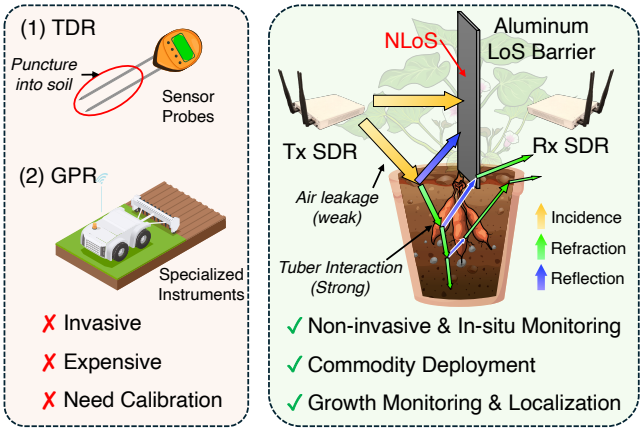



**Figure 1: Motivation for *SWAMP*. Unlike TDR and GPR, *SWAMP* uses commodity aboveground radios to enable non-invasive subsurface monitoring and localization.**

A natural question is whether *aboveground commodity wireless links* can provide useful information about belowground development. The hypothesis is physically plausible: radio propagation is a deterministic function of the medium. In soil, the effective complex permittivity and conductivity change with moisture, texture, ionic content, and the appearance of high-water-content biomass. These changes affect attenuation, phase delay, scattering, and multipath, suggesting that commodity radio-frequency (RF) measurements could encode signatures of belowground growth. However, realizing this idea in practice is challenging because soil is lossy, heterogeneous, and time-varying, and because commodity radio stacks were not designed for subsurface sensing.

Existing solutions leave a substantial deployability gap. Probe-based methods such as time-domain reflectometry (TDR) and frequency-domain reflectometry (FDR) can provide accurate point measurements but require insertion into soil and often need sensor- or soil-specific calibration [8,


*Conference'17, Washington, DC, USA*
2026. ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
https://doi.org/10.1145/nnnnnnn.nnnnnnn

22, 31]. Non-contact imaging approaches such as ground-penetrating radar (GPR) and electrical resistance tomography (ERT) can recover subsurface structure, but they often require dedicated survey hardware, trained operators, and soil-specific tuning, while performance can degrade in conductive or moisture-variable soils [20, 21, 42]. These constraints motivate a different operating point: a sensing system that avoids soil intrusion, relies on widely available radio frequency (RF) hardware and standardized metrics, and remains stable under realistic soil variability.

In this paper, we present *SWAMP* (**S**ubsurface **W**ireless **A**gricultural **M**onitoring **P**latform), a commodity-wireless platform for non-invasive subsurface monitoring and localization. *SWAMP* combines two complementary streams. First, a wideband software-defined radio (SDR) link performs repeatable channel frequency response (CFR) sweeps to capture spectral attenuation and spectral-structure changes induced by subsurface dielectric evolution. Second, a controllable Long Term Evolution (LTE) link collects standardized link-quality indicators—reference signal received power (RSRP), signal-to-interference-plus-noise ratio (SINR), modulation and coding scheme (MCS), block error rate (BLER), and throughput—whose spatial variation can be fused into coarse tuber probability maps and can also be inspected for auxiliary soil-condition studies. The two streams serve different but complementary roles: CFR sweeps support longitudinal monitoring at a fixed geometry, while LTE indicators provide spatial diversity for localization and controlled sensitivity tests.

Building such a system requires solving three technical challenges. **First**, soil variability induces channel changes unrelated to growth; without band selection and robust features, moisture-driven drift can dominate the signal of interest. **Second**, wideband CFR analysis benefits from repeatable complex spectra, but shared clocks and Global Positioning System-disciplined oscillators add substantial cost and deployment complexity. **Third**, localization from commodity link indicators is indirect because each metric is only a partial projection of the underlying channel, making multi-metric fusion necessary.

*SWAMP* addresses these issues using three design principles. (1) It treats **band selection** as an empirical design study rather than assuming a fixed best frequency a priori. (2) It uses a lightweight **Ethernet/TCP coordination protocol** together with per-tone carrier frequency offset (CFO) compensation and reference normalization to obtain repeatable wideband CFR sweeps without GPS-disciplined oscillator hardware. (3) It extracts four interpretable, band-aggregated descriptors for monitoring and combines standardized LTE metrics via a constrained fusion model for localization.

We evaluate *SWAMP* in a 45-day greenhouse sweet potato study spanning two soil recipes and two moisture regimes. Across these conditions, *SWAMP* reaches 83.0–91.0% accuracy for three-stage growth monitoring and 87.5–95.0% accuracy for 5 cm-grid localization. We additionally conduct auxiliary experiments on band selection, buried sweet potato count discrimination, and moisture/fertility sensitivity. The prototype costs roughly \$3k–\$4k, can be deployed in 10–20 minutes, and requires no buried sensors. Taken together, these results show that commodity wireless measurements contain reproducible belowground signatures under controlled but soil-diverse settings.

The contributions of this paper are:

- **A hybrid commodity-wireless architecture for subsurface crop sensing.** *SWAMP* combines repeatable wideband CFR sensing with standardized LTE link indicators to support longitudinal monitoring, coarse underground localization, and controlled sensitivity analysis.
- **Low-cost repeatable wideband acquisition without GPS-disciplined oscillators.** We introduce a lightweight Ethernet/TCP coordination mechanism with per-tone compensation and reference normalization that enables repeatable complex CFR sweeps suitable for wideband feature extraction.
- **Interpretable feature extraction and constrained multi-metric fusion.** We develop four CFR-derived features for growth monitoring and a nonnegative, simplex-constrained fusion method that combines five LTE indicators into tuber occupancy maps.
- **A reorganized greenhouse evaluation across five experimental tasks.** We validate *SWAMP* through band-selection analysis, underground localization, buried sweet potato count detection, moisture/fertility sensitivity tests, and longitudinal growth monitoring.

The remainder of this paper is organized as follows. Section 2 reviews the RF and soil-propagation background that motivates our design. Section 3 presents the overall system architecture, and Section 4 details the measurement and inference pipeline. Section 5 describes the prototype implementation and experimental setup. Section 6 reports results on band selection, underground localization, buried sweet potato count detection, moisture/fertility sensitivity, and longitudinal growth monitoring. Section 7 discusses limitations and implications for field-scale deployment, Section 8 summarizes related work, and Section 9 concludes.

## 2 Background and Design Implications

This section summarizes the RF background needed for *SWAMP* and makes the soil assumptions explicit, which is necessary to interpret wireless signatures in subsurface sensing.

## 2.1 RF propagation in soil

*2.1.1 Soil as a lossy, dispersive, and time-varying medium.* At RF frequencies, soil can be modeled by a frequency-dependent complex relative permittivity

$$\tilde{\epsilon}_r(f) = \epsilon_r'(f) - j\epsilon_r''(f), \tag{1}$$

where $\epsilon_r'(f)$ governs wave speed and refraction while $\epsilon_r''(f)$ captures dielectric loss. A common approximation links the loss term to effective conductivity $\sigma$:

$$\epsilon_r''(f) \approx \frac{\sigma}{2\pi f \epsilon_0}, \tag{2}$$

where $\epsilon_0$ is the vacuum permittivity. Increasing water content and ionic concentration typically change the effective dielectric response and bulk conductivity of soil, which jointly alter attenuation and phase delay [1, 7, 13, 18, 32, 33, 35].

In practice, soil is also heterogeneous and time-varying. Texture, compaction, pore distribution, and moisture can vary spatially, while irrigation and evaporation create day-to-day temporal drift. For sensing, this means that many observed channel changes are not caused by growth. The subsurface sensing problem is therefore fundamentally *confounded* by soil state. As a design-time approximation, *SWAMP* uses reference-normalized, shape-based features rather than relying on absolute calibration of the medium.

*2.1.2 Penetration versus sensitivity trade-off.* For a lossy dielectric, the propagation constant can be written as

$$\gamma(f) = \alpha(f) + j\beta(f), \tag{3}$$

where $\alpha(f)$ is the attenuation constant and $\beta(f)$ is the phase constant. A convenient penetration proxy is the power penetration depth $\delta_p(f) \approx 1/\alpha(f)$. Lower frequencies generally penetrate deeper, but shorter wavelengths at higher frequencies are more sensitive to smaller dielectric structures and produce richer frequency selectivity. This creates a basic trade-off: higher frequency and larger bandwidth improve sensitivity to fine structure, while lower frequency improves penetration in lossy soil. *SWAMP* therefore does not assume a universal best band across all settings.

*2.1.3 Interfaces, multipath, and spectral structure.* Even in a simple pot geometry, the effective channel is not single-path. The air–pot boundary, pot wall, soil volume, and tuber all produce reflections, scattering, and refraction. A useful abstraction is

$$H(f) = \sum_{k=1}^{K} a_k e^{-j2\pi f \tau_k}, \tag{4}$$

where each path $k$ has complex amplitude $a_k$ and delay $\tau_k$. Growth changes both the amplitudes and delays by introducing high-water-content biomass and additional scattering structure. Instead of attempting full inversion of this medium, *SWAMP* extracts robust descriptors of $H(f)$ that remain informative under moderate soil drift.

## 2.2 Why commodity wireless metrics matter

Existing subsurface sensing methods present clear deployment trade-offs. TDR and FDR probes can provide accurate point moisture sensing but require insertion into the medium and often need calibration that depends on sensor type, soil type, and operating conditions [8, 22, 31]. GPR and ERT can image or infer subsurface structure but often require more specialized survey instrumentation, inversion workflows, or both, and their performance is strongly coupled to soil conductivity, moisture, and texture [20, 21, 42]. By contrast, indicators such as RSRP, SINR, MCS, BLER, and throughput are standardized and widely available in cellular stacks. Designing around these existing metrics improves the path to deployability, even if the present prototype uses SDRs for control and repeatability.

## 2.3 Design implications

The physical background above leads to four system-level implications.

**Implication 1: band selection should be data driven.** A fixed one-size-fits-all frequency is unreliable in soil because attenuation and sensitivity shift with moisture and texture. We therefore present band selection as the first experimental study in Section 6.1 and then use the selected band consistently in later analyses.

**Implication 2: features must tolerate soil-driven drift.** Band-aggregated, shape-based metrics such as attenuation area, spectral slope, and ripple strength are more robust than single-tone amplitudes.

**Implication 3: monitoring and localization require different diversity.** Longitudinal monitoring benefits from a fixed geometry and a stable, interpretable probe signal. Localization requires spatial diversity and multi-metric evidence.

**Implication 4: standardized metrics improve deployability.** Cellular indicators such as RSRP, SINR, MCS, BLER, and throughput are widely exposed and reduce dependence on custom physical-layer modifications.

# 3 System Architecture

*SWAMP* targets a portable, non-invasive deployment model. A wideband SDR transmitter and receiver are positioned aboveground around the pot in a line-of-sight-suppressed geometry. A stepped-frequency sweep probes the medium across a wide band, and the receiver estimates a complex CFR vector after per-tone carrier-offset compensation. This sensing stream is designed for repeated daily measurements under a fixed geometry, enabling longitudinal monitoring.

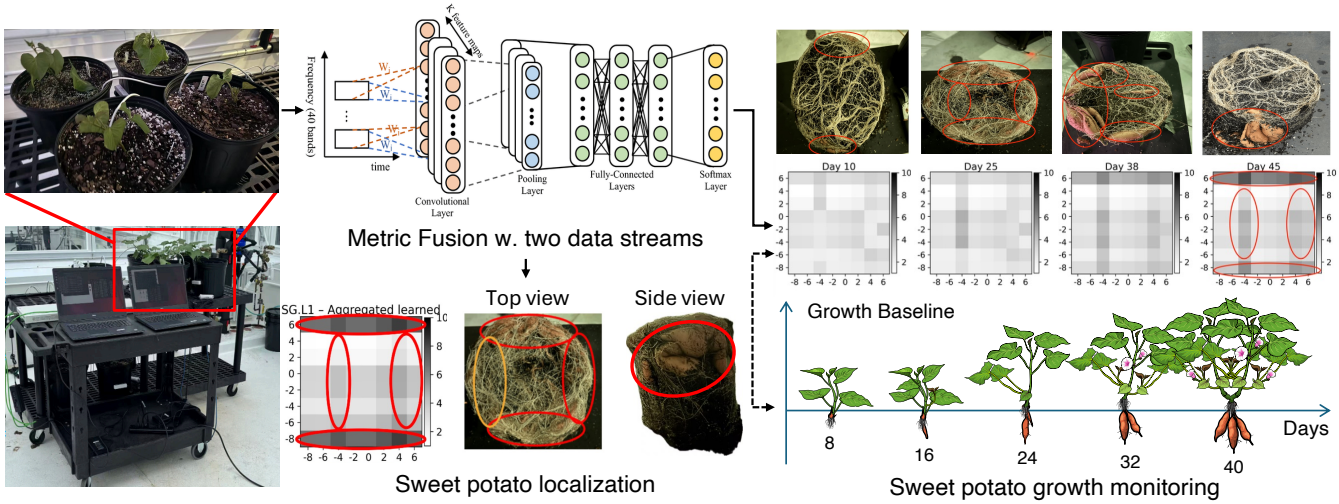



**Figure 2: Overview of *SWAMP*. A repeatable SDR-based stepped-frequency sweep provides complex CFR measurements for longitudinal monitoring, while a private LTE link produces standardized link-quality indicators for localization and auxiliary condition sensing.**

In parallel, *SWAMP* uses a private LTE link implemented with srsRAN and USRP X310 devices. During localization, the receiver side is raster-scanned over a 5 cm grid around the pot while the LTE stack logs standardized link indicators. The same LTE link is also reused for auxiliary controlled studies in which buried sweet potato count or soil condition is intentionally varied. Unlike the CFR mode, which uses a deliberately controlled probe signal, the LTE mode leverages link adaptation and multipath-sensitive metrics already provided by a commodity cellular stack.

## 3.1 Two sensing modes

*3.1.1 CFR mode for fixed-geometry monitoring.* For growth monitoring, *SWAMP* fixes the antenna geometry and performs one CFR sweep per day. The received complex gain across frequency is flattened by an air reference and normalized relative to day 1. Four interpretable descriptors summarize broadband attenuation, differential high-frequency loss, overall spectral tilt, and ripple strength. These descriptors are mapped to a three-stage growth estimate.

*3.1.2 LTE mode for localization and auxiliary sensing.* For localization, *SWAMP* raster-scans the receiver on a 5 cm grid and records LTE link indicators at each position. Each indicator yields a partial and noisy view of the channel. *SWAMP* therefore normalizes each metric per scan and fuses the resulting heatmaps through a constrained linear model. For controlled tuber-count and soil-condition studies, the same LTE indicators are aggregated per trial and compared across experimental conditions.

## 3.2 Design goals

*SWAMP* is guided by three practical goals. **First**, it must avoid disturbing the soil or embedding electronics. **Second**, it must remain low-cost and reproducible without demanding specialized hardware. **Third**, it must use features and metrics that are interpretable enough to understand when soil state, rather than growth, is driving the observed variation.

# 4 Measurement and Inference Design

## 4.1 Repeatable CFR acquisition

*4.1.1 Measurement geometry and line-of-sight suppression.* A major challenge in using aboveground wireless links for subsurface sensing is that direct air propagation can dominate the received signal and mask the soil-interacting component. To increase sensitivity to the pot and soil volume, *SWAMP* adopts a line-of-sight-suppressed geometry. The transmitter and receiver are placed on opposite sides of the pot, and an aluminum barrier blocks the dominant aboveground path. This does not eliminate all air-path leakage, but it shifts a larger fraction of received energy toward paths that interact with the pot, soil, and tuber volume. This operating principle is illustrated conceptually in Figure 1, while the full dataflow is shown in Figure 2.

This geometry is particularly important for longitudinal monitoring because it stabilizes the measurement semantics across days. By fixing antenna height, separation, and orientation, *SWAMP* reduces pose-induced variability and makes day-to-day differences more attributable to medium changes. The same geometric principle is also used in the LTE localization mode, where the receiver position changes

over a measurement grid while antenna polarization and height remain fixed.

*4.1.2 TCP-coordinated stepped-frequency sweep.* The transmitter and receiver hosts maintain a TCP control channel over Ethernet. For each sweep frequency $f_k$, the receiver first tunes to $f_k$ and reports readiness. The transmitter then tunes to $f_k$, waits for front-end settling, transmits a unit-amplitude tone, and signals the active capture window. The receiver records a short complex baseband segment and estimates the complex gain at that step. This deterministic sequencing reduces random timing variation across sweeps.

*4.1.3 Per-step CFO compensation and reference normalization.* For tone step $k$, the received baseband can be approximated as

$$r_k(t) = H(f_k)e^{j2\pi\Delta f_k t} + n(t), \tag{5}$$

where $\Delta f_k$ is residual carrier frequency offset and $n(t)$ is noise plus interference. We estimate $\widehat{\Delta f_k}$ by fitting a line to the unwrapped phase over time and then compute

$$\widehat{H}(f_k) = \frac{1}{T}\int_0^T r_k(t)e^{-j2\pi\widehat{\Delta f_k}t}\,dt. \tag{6}$$

Although this does not produce globally phase-absolute measurements, it yields repeatable relative complex gains across the sweep that are sufficient for robust feature extraction.

To remove static front-end ripple and frequency-dependent antenna response, *SWAMP* records an air reference sweep with the same antenna separation but no pot between the antennas. Let $\widehat{H}_d^{\text{pot}}(f)$ be the raw sweep for a pot on day $d$ and $\widehat{H}^{\text{air}}(f)$ the air reference. The flattened CFR is

$$\widehat{H}_d^{\text{flat}}(f) = \frac{\widehat{H}_d^{\text{pot}}(f)}{\widehat{H}^{\text{air}}(f)}. \tag{7}$$

For longitudinal analysis, each day is then normalized relative to day 1 for the same pot:

$$\widehat{H}_d^{\text{norm}}(f) = \frac{\widehat{H}_d^{\text{flat}}(f)}{\widehat{H}_1^{\text{flat}}(f)}. \tag{8}$$

This suppresses fixed hardware artifacts and emphasizes relative evolution of the soil-tuber system.

## 4.2 Feature design for growth monitoring

*4.2.1 Working-band principle.* Because soil attenuation and sensitivity vary strongly with frequency, *SWAMP* does not assume that the full swept band is equally useful. Instead, it first performs an empirical band-selection study by jointly considering (i) the contrast between soil-only and soil-plus-tuber measurements, (ii) usable signal-to-noise ratio (SNR) under the tested soil and moisture conditions, and (iii) stability across repeated measurements. The selected working band used throughout this paper is reported in Section 6.1.

**Algorithm 1** Repeatable stepped-frequency CFR acquisition

1: **for** frequency step $f_k \in \{f_1, \ldots, f_N\}$ **do**
2: RX tunes to $f_k$ and sends `READY`
3: TX tunes to $f_k$ and waits for RF settling
4: TX transmits a unit-amplitude tone
5: RX records complex baseband samples for duration $T$
6: Estimate residual CFO $\widehat{\Delta f_k}$ from phase slope
7: Compute $\widehat{H}(f_k)$ by CFO-compensated averaging
8: Flatten sweep by air reference and normalize to day 1
9: Extract CFR features on the selected working band

This organization keeps the design principle in the method section while placing the actual band choice in the experimental section where the evidence is shown.

Within the selected band, *SWAMP* summarizes each normalized CFR using four interpretable features derived from the magnitude spectrum. Let

$$A_{\text{dB}}(f) = 20\log_{10}\left|\widehat{H}^{\text{norm}}(f)\right|. \tag{9}$$

*4.2.2 Selected magnitude-domain features.*

*Broadband Attenuation Integral (BAI)..* BAI captures aggregate attenuation area across the band:

$$\text{BAI} = \frac{1}{f_2 - f_1}\int_{f_1}^{f_2}\left(A_{\max} - A_{\text{dB}}(f)\right)\,df, \quad A_{\max} = \max_{f\in[f_1,f_2]} A_{\text{dB}}(f). \tag{10}$$

*Spectral slope.* A least-squares line is fit to $A_{\text{dB}}(f)$ over the working band. The slope captures frequency-dependent attenuation, which tends to become more negative as the medium becomes more lossy.

*High/Low band ratio (H/L)..* Let $f_m = (f_1 + f_2)/2$. We compute

$$\text{H/L} = \frac{\int_{f_m}^{f_2}\left|\widehat{H}^{\text{norm}}(f)\right|\,df}{\int_{f_1}^{f_m}\left|\widehat{H}^{\text{norm}}(f)\right|\,df}. \tag{11}$$

As higher-frequency attenuation increases, this ratio decreases.

*Ripple variance.* A smoothed baseline $\widetilde{A}(f)$ is computed from the linear magnitude spectrum using median filtering, and the normalized ripple strength is measured as

$$\text{RippleVar} = \text{std}\left(\frac{\left|\widehat{H}^{\text{norm}}(f)\right|}{\widetilde{A}(f)}\right). \tag{12}$$

This feature captures fine-scale spectral variation produced by additional scattering interfaces and boundary-coupled multipath.



**(a) Greenhouse deployment at the start of the study.** **(b) Mature canopy later in the 45-day experiment.** **(c) Intact harvested root ball.** **(d) Top view of the tuber arrangement.**

**Figure 3: Greenhouse deployment photographs.**

These four features are intentionally simple. They summarize broad physical trends rather than overfitting the detailed shape of the spectrum, which is critical given the small number of biologically independent pots in the current dataset.

*4.2.3 Growth-stage inference.* *SWAMP* formulates longitudinal monitoring as a three-stage classification problem. Since direct non-destructive tuber labels are unavailable for each day, the current study uses day index as a proxy for growth stage: early (days 1–15), middle (days 16–30), and late (days 31–45). We emphasize that these are *time bins*, not independently validated phenological stages. Sweet potato phenology can also be described more formally using extended BBCH-style stages [28], but the current dataset does not include repeated destructive or imaging labels needed to map each day to those formal stages. The resulting four-dimensional feature vector is standardized using training-set statistics and passed to a multinomial logistic regression classifier with L2 regularization. This lightweight model matches the scale of the available dataset and supports leave-one-pot-out evaluation.

## 4.3 LTE metrics and fusion

*4.3.1 Logged LTE indicators.* CFR sweeps are effective for fixed-geometry monitoring, but localization requires spatial diversity. *SWAMP* therefore uses a controllable private LTE link because LTE stacks naturally expose standardized indicators that summarize different aspects of link quality. At each grid cell or auxiliary trial, the system records:

- **RSRP**: reference signal received power;
- **SINR**: signal-to-interference-plus-noise ratio;
- **MCS**: modulation and coding scheme index;
- **BLER**: block error rate;
- **Throughput**: achieved data rate.

No single metric is consistently reliable across all tested soil and moisture conditions. This is expected: each indicator compresses the same underlying channel into a different control-plane summary. *SWAMP* therefore treats these quantities as complementary views and fuses them into a single spatial score map for localization. For the auxiliary tuber-count and soil-condition experiments, the same metrics are compared directly across controlled conditions rather than fused into a grid map.

*4.3.2 Constrained multi-metric fusion.* Let $g$ index grid cells and let $x_j(g)$ denote normalized LTE metric $j$ at cell $g$. *SWAMP* predicts a tuber occupancy score using a constrained linear model:

$$\widehat{y}(g) = \sum_{j=1}^{5} w_j x_j(g), \quad w_j \geq 0, \quad \sum_{j=1}^{5} w_j = 1. \tag{13}$$

The nonnegativity and simplex constraints make the fusion easy to interpret and prevent unstable cancellation between metrics. We parameterize the weights with a softmax:

$$w_j = \frac{\exp(u_j)}{\sum_{k=1}^{5} \exp(u_k)}, \tag{14}$$

and learn $u$ by minimizing mean squared error (MSE) to the ground-truth occupancy map over training pots. Optimization is performed with Broyden–Fletcher–Goldfarb–Shanno (BFGS). For visualization, the resulting score map is rescaled to $[0, 1]$ per pot.

## 5 Prototype and Experimental Setup

*SWAMP* is implemented using commodity SDR hardware and open-source software. Two USRP X310 radios form the CFR link, while another X310 pair supports the LTE evolved NodeB (eNodeB) and user equipment (UE) through srsRAN. All components are controlled from commodity laptops connected over Ethernet. The study was conducted within NC State controlled-environment plant research infrastructure, including environmentally controlled greenhouses and dedicated plant growth facilities [23, 24]. Figures 2 and 3 show

**Algorithm 2** *SWAMP* localization fusion

**Require:** Grid cells $\mathcal{G}$, normalized LTE metrics $\{x_j(g)\}_{j=1}^{5}$, training labels $y(g)$

Initialize unconstrained parameters $u_j \leftarrow 0$
**repeat**
  Convert weights: $w_j \leftarrow \exp(u_j)/\sum_k \exp(u_k)$
  Compute prediction: $\widehat{y}(g) \leftarrow \sum_j w_j x_j(g)$ for all $g \in \mathcal{G}$
  Update $u$ using BFGS on loss $\mathcal{L} = \frac{1}{|\mathcal{G}|}\sum_g(\widehat{y}(g) - y(g))^2$
**until** convergence
Rescale $\widehat{y}(g)$ to $[0, 1]$ per pot for visualization
**return** tuber probability map $\widehat{y}(g)$

the greenhouse deployment, representative canopy growth, and representative harvest views used to interpret ground truth.

## 5.1 Prototype implementation

*5.1.1 CFR link.* The stepped-frequency CFR link is built with two USRP X310 devices controlled through GNU Radio. The transmitter emits a unit-amplitude sinusoidal tone and steps from 2.0 GHz to 5.0 GHz in 40 MHz increments. At each frequency, the system waits 0.3 s for front-end settling and then records 0.2 s of complex baseband at 500 kS/s. The local oscillator (LO) is set to the instantaneous center frequency at each step.

*5.1.2 LTE link.* The localization and auxiliary-condition link is implemented with srsRAN 4G. The eNodeB and UE run on commodity laptops, each connected to a USRP X310. The LTE cell operates at 2680 MHz with $n_{\mathrm{PRB}} = 50$ (10 MHz bandwidth), where PRB denotes physical resource block, and a 1.92 MS/s baseband sampling rate. At each grid position, the system maintains the link for 20 s and records link indicators at 5 Hz, yielding 100 raw samples per metric per position. Transient samples immediately after motion or reattachment are discarded, and the remaining samples are aggregated using the median.

*5.1.3 Antenna geometry and repeatable placement.* National Instruments (NI) VERT2450 antennas are used across the SDR links. The antennas are mounted 8 in above the floor, the pot diameter is 8 in, and the antenna separation is 10 in. A 10 in aluminum barrier is placed between the antennas to suppress the direct path. The pot and radio carts are marked so that the relative geometry can be reproduced across daily measurements. The deployment shown in Figure 3a is used for fixed-geometry CFR monitoring, while the harvest views in Figure 2 provide qualitative top-view and side-view context for the final tuber span used to interpret localization.

**Table 1: Key implementation parameters used in the current *SWAMP* prototype.**

| Component | Setting |
|---|---|
| CFR radios | USRP X310 TX/RX pair |
| CFR software | GNU Radio |
| CFR sweep range | 2.0–5.0 GHz in 40 MHz steps |
| CFR working band | 2.0–3.5 GHz (selected in Sec. 6.1) |
| CFR settle / record window | 0.3 s / 0.2 s per step |
| CFR baseband sample rate | 500 kS/s |
| LTE radios | USRP X310 eNodeB / UE pair |
| LTE software | srsRAN 4G |
| LTE center frequency | 2680 MHz |
| LTE bandwidth | 10 MHz ($n_{\mathrm{PRB}} = 50$) |
| LTE sampling window | 20 s per grid cell at 5 Hz metrics |
| Grid resolution | 5 cm |
| Antenna separation | 10 in |
| Antenna height | 8 in |
| Line-of-sight suppression | 10 in aluminum barrier |
| System cost | \$3,000–\$4,000 |
| Deployment time | 10–20 min |

## 5.2 Datasets and greenhouse deployment

*5.2.1 Main 45-day growth study.* We evaluate *SWAMP* using four 3-gallon sweet potato pots, each representing one combination of soil recipe and watering regime: SG.L1, SG.L2, CB.L1, and CB.L2. The two soil recipes are Sunshine Propagation Mix (SG) and Jolly Gardener HFCB Mix (CB). These choices are consistent with controlled-environment container practice, where soilless or semi-soilless substrates are selected for tunable water-holding capacity, aeration, pH/EC management, and repeatability across pots [2, 14, 15, 27, 29, 39]. Controlled-environment facilities such as the NC State Phytotron also document standard substrate recipes based on propagation mix blended with mineral fractions for repeatable experiments [25]. The two watering regimes are denoted L1 and L2 and are implemented through distinct watering patterns. One CFR sweep is collected per day per pot at a fixed time and fixed geometry. For localization, LTE scans are performed on a 5 cm grid and the inferred maps are compared to tuber ground truth annotated at harvest.

*5.2.2 Auxiliary controlled studies.* Beyond the four main growth pots, we include additional controlled perturbation experiments to test what the wireless channel is sensitive to. These auxiliary studies vary buried sweet potato amount and physical state, as well as soil moisture and fertility conditions. They are not treated as additional biological replicates for growth-stage classification; instead, they are used to examine sensitivity, confounding, and the separability of different underground conditions.

**Table 2: Representative compartment ranges from NC State Phytotron status board, providing facility-level environmental context.**

| Parameter | Value range |
|---|---|
| Temperature setpoint across G1–G10 | 21.5–29.8 °C |
| Actual temperature across G1–G10 | 20.6–28.4 °C |
| $CO_2$ setpoint across G1–G10 | 350–400 ppm |
| Actual $CO_2$ across G1–G10 | 382–605 ppm |
| Relative humidity setpoint across G1–G10 | 47.5–60.0% |
| Actual relative humidity across G1–G10 | 45.6–65.0% |
| PAR across G1–G10 | 4–489 $\mu$mol/m$^2$/s |

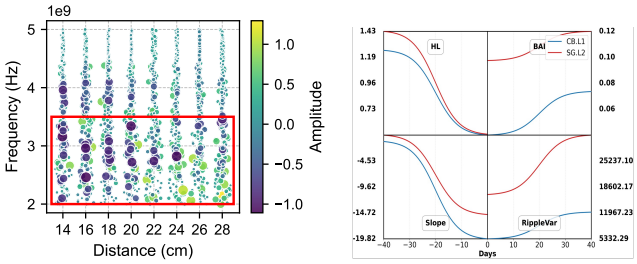
*5.2.3 Environmental control context.* The greenhouse environment is intentionally controlled to isolate the core sensing problem. Temperature and humidity are kept approximately constant, and the primary experimental factors are soil recipe and moisture distribution. This controlled-environment framing is consistent with the documented capabilities of the NC State Phytotron and associated Plant Sciences Building greenhouses [24, 26]. The current dataset contains only four biologically independent pots. Daily sweeps and repeated LTE samples per grid cell are technical replicates used to reduce measurement noise, not additional biological replicates. We therefore treat the study as a controlled proof-of-concept and use lightweight models throughout.

## 5.3 Parameters and evaluation metrics

*5.3.1 System parameters.*

*5.3.2 Representative greenhouse parameters from the attached status board.* Table 2 summarizes representative Phytotron board-reported compartment ranges visible in the attached greenhouse status photo. We include these values as environmental context rather than as exact per-pot labels. They document the kinds of temperature, carbon dioxide ($CO_2$), humidity, and photosynthetically active radiation (PAR) settings present in the controlled facility during the campaign, and they are broadly consistent with the operating envelopes documented for the facility's greenhouse infrastructure [24, 26].

*5.3.3 Evaluation metrics.* For growth monitoring, we report three-stage classification accuracy using leave-one-pot-out evaluation. For localization, we compare predicted maps with ground-truth occupancy maps using thresholded accuracy under tolerance margins (5 cm and 10 cm), mean squared error (MSE), and structural similarity index (SSIM). Tolerance-dilated accuracy is included because sparse occupancy maps are sensitive to small annotation shifts. For the auxiliary tuber-count and soil-condition studies, we mainly examine whether aggregated metric signatures separate experimental conditions in a stable and interpretable way.




**(a) Band selection from CFR sweeps.** **(b) Growth monitoring with CFR features.**

**Figure 4: CFR-based results.**

# 6 Experimental Results

We organize the experiments in the same order that they support the overall system pipeline. We first identify the usable CFR working band. We then evaluate underground localization, study sensitivity to buried sweet potato count and soil condition, and finally show that the four selected CFR-derived indicators support longitudinal growth monitoring. Figures 4 and 5 summarize the CFR-based and localization-oriented outputs used throughout the discussion. Figure 6 groups the auxiliary controlled studies, and Figure 7 visualizes the qualitative spatial evolution of inferred occupancy across time for two representative pots.

## 6.1 Band selection study

Figure 4a illustrates representative CFR magnitude patterns across frequency. The most stable contrast between soil-only and soil-plus-tuber cases concentrates in the 2.0–3.5 GHz region, with a prominent absorption trough near 2.68 GHz in the current setup. Above roughly 3.5 GHz, attenuation increases while the incremental contrast becomes less consistent across pots and moisture conditions. This observation motivates the use of 2.0–3.5 GHz as the working band for all downstream CFR features.

The selected band also aligns with the multilayer interpretation in Section 2.1: lower frequencies provide better penetration but weaker sensitivity to localized dielectric discontinuities, whereas higher frequencies offer richer spectral activity but attenuate more rapidly in wet, conductive soil. In our pot geometry, the mid-gigahertz band provides the best compromise. For this reason, band selection is best presented as the first experiment rather than as a fixed assumption inside the design section.

## 6.2 Localization with LTE metric fusion

We next evaluate grid-level localization using LTE-derived metrics. Representative single-metric responses are shown in Figure 5, while representative harvest views of the final root-ball topology are shown in Figure 3. Along the lateral

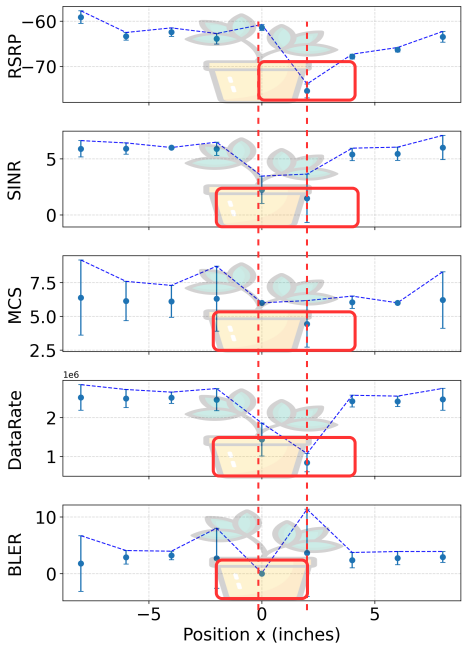



**Figure 5: Representative per-metric LTE responses along the lateral scan axis. Each subplot shows one standardized link indicator.**

scan axis, individual indicators such as RSRP, SINR, MCS, throughput, and BLER all exhibit deviations over the ground-truth tuber span, but the deviations are not perfectly aligned in amplitude or position. In the current data, some metrics degrade strongly in one condition and weakly in another. This confirms that no single LTE indicator is a reliable stand-in for underground occupancy.

The constrained fusion model improves both discrete and continuous agreement with ground truth. On the current dataset, the fused map reaches 87.5–95.0% localization accuracy on a 5 cm grid, outperforming single-metric baselines. Table 3 summarizes representative map-level scores from the reference analysis: the fused map reduces MSE to 0.115, compared with 0.122–0.239 for single-metric heatmaps, while the corresponding SSIM value is 0.262. This gain is important because the map is sparse: many cells are empty, and naive thresholding can appear deceptively accurate without preserving the structure of the occupied region. The fusion model improves not only cell-wise agreement but also the overall shape of the predicted tuber region.

Qualitatively, the learned weights tend to balance power-related metrics with reliability-related metrics. RSRP and throughput capture coarse signal-strength changes, while BLER and SINR become useful when link adaptation amplifies small channel changes into decoding outcomes. MCS acts as a compact summary of channel quality after adaptation. The simplex-constrained model cannot represent complex

**Table 3: Representative map-level localization scores for single LTE metrics and the fused map.**

| Metric | BLER | Rate | SINR | RSRP | Fused |
|---|---|---|---|---|---|
| MSE | 0.239 | 0.163 | 0.137 | 0.122 | **0.115** |
| SSIM | 0.337 | 0.315 | 0.320 | 0.264 | **0.262** |

nonlinear interactions, but in this small-data regime it provides a stable and interpretable baseline.

## 6.3 Buried sweet potato count detection

We next ask whether the wireless stream is sensitive not only to presence or absence, but also to the amount of buried sweet potato material. The controlled count/amount experiment in Figure 6c varies the number or effective amount of sweet potato buried in the soil while keeping the overall measurement procedure fixed. Across these trials, aggregated LTE metrics show separable signatures rather than collapsing into a single occupied-state response. In particular, cases with more buried sweet potato mass tend to produce stronger metric shifts than smaller-count or partial-burial cases.

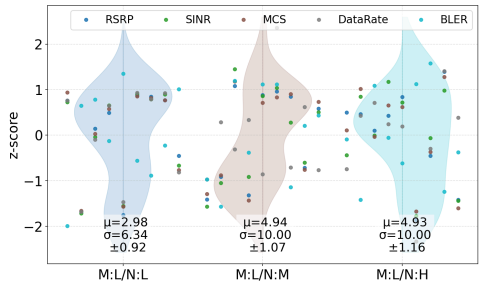
We interpret this result cautiously. The observed differences likely reflect a mixture of biomass, water distribution, scattering structure, burial depth, and coupling to the pot boundary. Accordingly, this experiment should be read as a sensitivity study rather than as a finalized count classifier. Still, it indicates that the wireless channel is responsive to more than a binary occupancy label, which supports future work on tuber-count or tuber-mass estimation.

## 6.4 Moisture and fertility sensitivity

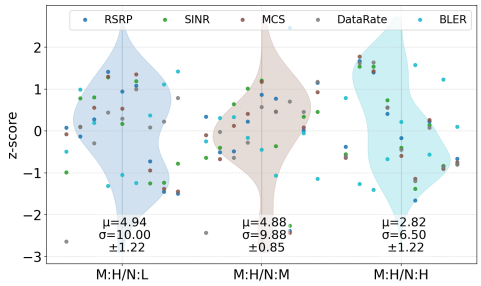
The moisture/fertility experiment examines a key confounder for wireless underground sensing. This concern is agronomically meaningful in sweet potato because early-season soil moisture regimes affect storage-root initiation, growth, and physiology even under greenhouse conditions [11, 12]. Figures 6a and 6b show that z-score-normalized LTE metrics shift systematically when soil moisture and fertility conditions are changed under otherwise controlled settings. This is consistent with the RF background in Section 2.1: conductivity and effective permittivity alter attenuation and multipath even when tuber geometry is fixed. This finding is important for interpretation. A model that ignores moisture or fertility may attribute soil-state changes to biological growth.

## 6.5 Growth monitoring with CFR indicators

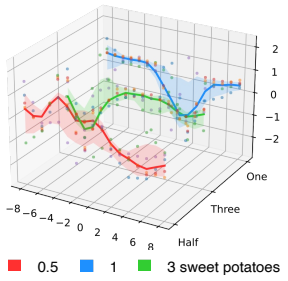
We finally evaluate whether fixed-geometry CFR sweeps contain repeatable temporal signatures of tuber development. The temporal evolution is visualized in Figure 4b, while Figure 7 provides qualitative examples of how inferred occupancy maps sharpen over time for two representative pots.




(a) Low-moisture vs. three nitrogen levels.

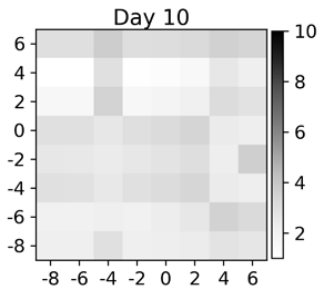



(b) High-moisture vs. three nitrogen levels.

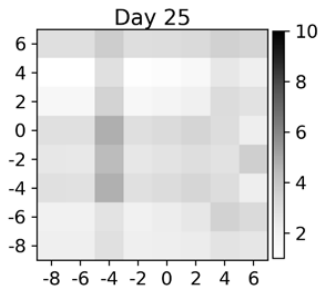



(c) Sweet potato count.

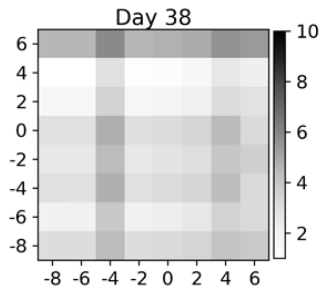
**Figure 6: Auxiliary controlled studies using LTE-derived metrics.**



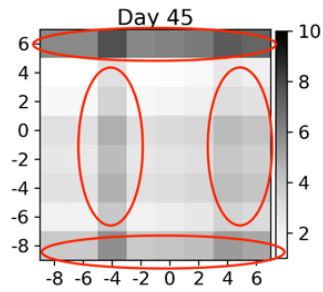

(a) SG.L1, day 10.



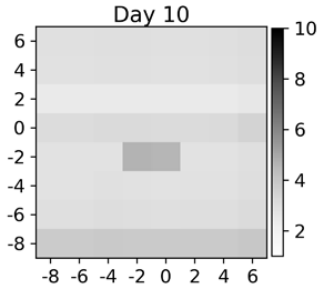

(b) SG.L1, day 25.



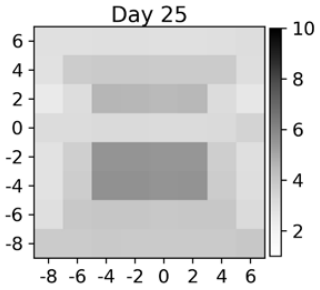

(c) SG.L1, day 38.



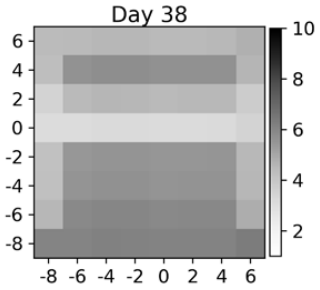

(d) SG.L1, day 45.



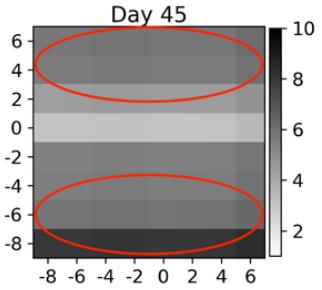

(e) CB.L1, day 10.




(f) CB.L1, day 25.




(g) CB.L1, day 38.




(h) CB.L1, day 45.

**Figure 7: Qualitative evolution of inferred tuber occupancy maps across time for two representative pots. From day 10 to day 45, the occupied region becomes more structured and spatially concentrated.**

As tubers develop and the effective water-rich biomass increases, broadband attenuation rises, making BAI larger. Because higher-frequency components attenuate more strongly, the H/L ratio decreases over time and the spectral slope becomes more negative. Ripple variance increases as the underground structure introduces additional scattering and boundary-coupled paths.

These trajectories are not perfectly monotonic on every day, which is expected given day-to-day moisture fluctuations and the small number of pots. Nonetheless, the overall trends remain reproducible enough to support a three-stage classification task. Using logistic regression with leave-one-pot-out evaluation, *SWAMP* achieves 83.0–91.0% accuracy for three-stage growth monitoring on the current dataset. Because the biologically independent sample size is small, we interpret these numbers as evidence of consistent information content rather than as a final estimate of generalization.

Feature-level behavior is also useful qualitatively. BAI and H/L capture coarse, physically interpretable loss trends. Spectral slope provides a compact summary of differential attenuation across the band. RippleVar becomes more pronounced when the underground structure is large enough to create spectral ripples. Together, these descriptors provide a robust compromise between interpretability and expressiveness.

## 6.6 Cost and deployment comparison

Table 4 summarizes a deployment-oriented comparison between *SWAMP* and representative subsurface sensing approaches. The goal is not to claim superiority on every axis, but to clarify the operating point of the system. *SWAMP*

**Table 4: Deployment-oriented comparison between *SWAMP* and representative subsurface sensing modalities.**

| Aspect | *SWAMP* | TDR probes | FDR probes | GPR / ERT |
|---|---|---|---|---|
| Primary task | Tuber monitoring and coarse localization | Point soil moisture sensing | Point soil moisture sensing | Subsurface imaging / conductivity mapping |
| Intrusiveness | Non-invasive; antennas remain aboveground | Probe insertion required | Probe insertion required | Non-invasive but requires dedicated instrumentation |
| System cost | $3,000–$4,000 depending on SDR reuse | $5,000–$10,000 with logger and probes | $200+ per probe plus reader electronics | $15,000–$50,000 or more |
| Setup time | 10–20 min in current geometry | 30–60 min with insertion and calibration | 30–60 min with insertion and calibration | 1–3 h including survey setup and parameter tuning |
| Spatial footprint | Repositionable scan with same hardware | One point per probe | One point per probe | Area survey over grids |
| Maintenance | Low; mostly recalibration and alignment checks | Probe wear, recalibration, replacement | Probe wear, recalibration, replacement | Moderate; antenna and calibration upkeep |
| Performance | 83.0–91.0% stage accuracy; 87.5–95.0% localization on 5 cm grid | High point accuracy after calibration [22, 31] | Sensor- and soil-dependent; calibration improves accuracy [22, 31] | Soil-dependent; penetration quality varies [20, 21, 42] |

avoids soil insertion, uses aboveground radios, requires relatively modest setup time, and supports both monitoring and localization in a single platform. It does not yet replace calibrated point moisture probes or dedicated subsurface imaging systems in their strongest regimes; rather, it offers a proof-of-concept for a different point in the design space.

## 7 Discussion

In this section, we discuss the current results: aboveground commodity radio links for belowground tuber sensing. We also make clear what *SWAMP* does *not* yet solve.

### 7.1 Interpretation and limitations

*7.1.1 Soil state remains the dominant confounder.* Moisture, conductivity, and nutrient-dependent ionic content can reshape the propagation channel even in the absence of biological growth. This is visible both in the main longitudinal data and in the dedicated moisture/fertility experiment in Section 6.4. The immediate implication is that future deployments cannot rely on raw wireless metrics alone. Robust sensing will likely require either condition-aware models, additional environmental inputs, or explicit calibration strategies that separate soil state from growth state.

*7.1.2 The current evidence is pot-scale and proof-of-concept.* The experiments involve four biologically independent pots in a greenhouse and intentionally controlled geometry. This is enough to demonstrate signal existence and to compare sensing modalities, but not enough to establish broad generalization across cultivars, field soil volumes, canopy conditions, or weather-driven moisture variation. The present accuracy numbers should therefore be interpreted as proof-of-concept results rather than final field-level performance.

*7.1.3 Fixed geometry helps monitoring but limits generality.* For the monitoring task, repeatability is achieved partly by fixing the antenna geometry and suppressing the direct path. This is a reasonable choice in a controlled study, but field deployments will introduce pose variation, canopy growth, and richer multipath. One likely path forward is to use more geometric diversity rather than less: multiple viewpoints, multiple antenna baselines, or repeatable mechanical scanning may improve robustness while preserving non-invasiveness.

*7.1.4 Commodity metrics as a feature, not a weakness.* A notable design choice in *SWAMP* is the use of standardized LTE indicators for localization and auxiliary sensing. These metrics are indirect and lossy summaries of the channel, but they are also exactly the kind of information exposed by commodity systems. This suggests a path toward eventual deployment on more practical hardware than SDRs, provided that the calibration and robustness problems can be addressed. In that sense, the present SDR-based prototype should be viewed as a controlled bridge between communication hardware and deployable sensing.

### 7.2 Toward field-scale deployment

A practical transition beyond greenhouse pots will likely require three extensions. First, systems should explicitly model or estimate soil state to separate growth-induced changes from moisture-driven drift. Second, sensing geometry should become more robust through additional spatial diversity, multiple links, or self-calibrating placement aids. Third, validation should be anchored to richer biological ground truth,

such as staged destructive sampling or complementary imaging, rather than day-indexed proxies alone.

## 8 Related Work

We position *SWAMP* relative to four research threads: root phenotyping and underground crop observation, electromagnetic soil sensing, commodity RF sensing for agriculture, and wideband wireless sensing systems.

### 8.1 Root phenotyping and underground crop observation

Root systems and storage organs are difficult to observe in situ, which has driven extensive work on excavation-based phenotyping, root observation platforms, and non-destructive underground monitoring [19, 30, 38]. Recent systems include integrated in situ root phenotyping platforms such as RhizoPot and distributed fiber-optic monitoring for underground root development [34, 41]. For storage organs specifically, developmental reviews and phenological staging work in sweet potato underline both the agronomic importance of hidden organs and the difficulty of obtaining repeated labels without destructive inspection [28, 44]. *SWAMP* differs from these approaches by using aboveground wireless links rather than transparent growth containers, optical access, or dedicated fiber installations.

### 8.2 Electromagnetic sensing of soil

A large body of work studies electromagnetic approaches for soil water and root-zone characterization. Reviews of TDR/FDR and related dielectric sensors show that these methods can provide accurate point measurements, but they rely on direct coupling to the medium, offer limited spatial footprint unless many probes are deployed, and often require calibration procedures that depend on sensor type, soil type, and operating conditions [1, 8, 22, 31]. In parallel, root-zone and subsurface imaging have been explored using electrical imaging/ERT and GPR. These methods can recover spatial structure without destructive excavation, but the pipeline typically depends on hardware, inversion or specialized post-processing, and performance that varies with conductivity, moisture, texture, and survey geometry [20, 21, 42]. *SWAMP* targets a different operating point: aboveground communication hardware, no soil insertion, and lightweight feature extraction rather than geophysical inversion.

### 8.3 RF sensing with commodity and communication-oriented signals

Recent systems work has shown that communication-centric RF measurements can support environmental and agricultural sensing tasks. LTE-based sensing has been used both as a general sensing substrate and as a low-cost soil moisture sensor, while other systems exploit LoRa links, RF paths, UAV-mounted IR-UWB radar, or battery-free chipless tags to infer soil moisture without dense wired instrumentation [3, 6, 9, 10, 16, 17]. *SWAMP* builds on the same broad insight that commodity wireless measurements can capture agriculturally meaningful changes, but differs in both target and architecture. First, it focuses on *subsurface crop development* rather than soil moisture estimation alone. Second, it combines a repeatable wideband CFR mode for longitudinal monitoring with standardized LTE link indicators for localization, rather than relying on a single RF modality.

### 8.4 Wideband and mobile sensing

A separate line of work in top networking and sensing venues studies how bandwidth, mobility, and spatial diversity improve RF sensing fidelity. WideSee demonstrates contactless sensing with LoRa, Octopus provides a practical wideband MIMO platform for higher-resolution wireless sensing, and Mobi$^2$Sense shows how controlled device motion can expand sensing capability beyond static links [4, 5, 40]. Systems such as SiWa further show that wideband RF can recover structure hidden behind visually opaque media when paired with task-specific inference pipelines [43]. Most closely related are recent root-tuber RF network systems that surround the target volume with many wireless links and reconstruct underground structure using learned tomographic models [36, 37].

Prior work shows that RF measurements can support soil sensing, but many of which rely on intrusive probes and denser infrastructure. *SWAMP* instead targets an aboveground, low-cost deployment model that combines repeatable wideband monitoring with standardized LTE metrics.

## 9 Conclusion

This paper presents *SWAMP*, a commodity-RF platform for non-invasive subsurface agricultural sensing. *SWAMP* combines repeatable wideband CFR sweeps for longitudinal monitoring with LTE link-quality indicators for localization and controlled sensitivity analysis. It uses a lightweight TCP-coordinated sweep protocol with per-step CFO compensation and reference normalization, together with four CFR features and a constrained LTE fusion model. In a 45-day greenhouse study spanning two soil recipes and two moisture regimes, *SWAMP* achieved 83.0–91.0% accuracy for three-stage growth monitoring and 87.5–95.0% accuracy for 5 cm-grid localization. Additional controlled experiments showed sensitivity to buried sweet potato amount as well as to soil moisture and fertility, highlighting both the utility of the wireless signals and the role of soil-state confounding. Overall, *SWAMP* demonstrates that commodity aboveground wireless links can provide a practical and scalable foundation for non-invasive monitoring of belowground crop development.

## References


[1] Mukhtar Iderawumi Abdulraheem, Hongjun Chen, Linze Li, Abiodun Yusuff Moshood, Wei Zhang, Yani Xiong, Yanyan Zhang, Lateef Bamidele Taiwo, Aitazaz A. Farooque, and Jiandong Hu. 2024. Recent Advances in Dielectric Properties-Based Soil Water Content Measurements. *Remote Sensing* 16, 8 (2024), 1328. doi:10.3390/rs16081328

[2] Ted E. Bilderback, Stuart L. Warren, James S. Owen, Jr., and Joseph P. Albano. 2005. Healthy Substrates Need Physicals Too! *HortTechnology* 15, 4 (2005), 747–751. doi:10.21273/HORTTECH.15.4.0747

[3] Zhaoxin Chang, Fusang Zhang, Jie Xiong, Junqi Ma, Beihong Jin, and Daqing Zhang. 2022. Sensor-Free Soil Moisture Sensing Using LoRa Signals. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 6, 2 (2022), 45:1–45:27. doi:10.1145/3534608

[4] Lili Chen, Jie Xiong, Xiaojiang Chen, Sunghoon Ivan Lee, Kai Chen, Dianhe Han, Dingyi Fang, Zhanyong Tang, and Zheng Wang. 2019. WideSee: Towards Wide-Area Contactless Wireless Sensing. In *Proceedings of the 17th ACM Conference on Embedded Networked Sensor Systems*. 258–270. doi:10.1145/3356250.3360031

[5] Zhe Chen, Tianyue Zheng, and Jun Luo. 2021. Octopus: A Practical and Versatile Wideband MIMO Sensing Platform. In *Proceedings of the 27th Annual International Conference on Mobile Computing and Networking*. 601–614. doi:10.1145/3447993.3483267

[6] Rong Ding, Haiming Jin, Dong Xiang, Xiaocheng Wang, Yongkui Zhang, Dingman Shen, Lu Su, Wentian Hao, Mingyuan Tao, Xinbing Wang, and Chenghu Zhou. 2023. Soil Moisture Sensing with UAV-Mounted IR-UWB Radar and Deep Learning. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 7, 1 (2023), 11:1–11:25. doi:10.1145/3580867

[7] Myron C. Dobson, Fawwaz T. Ulaby, Martti T. Hallikainen, and Mohamed A. El-Rayes. 1985. Microwave dielectric behavior of wet soil—Part II: Dielectric mixing models. *IEEE Transactions on Geoscience and Remote Sensing* GE-23, 1 (1985), 35–46. doi:10.1109/TGRS.1985.289498

[8] Steven R. Evett and Giles W. Parkin. 2005. Advances in Soil Water Content Sensing: The Continuing Maturation of Technology and Theory. *Vadose Zone Journal* 4, 4 (2005), 986–991. doi:10.2136/vzj2005.0099

[9] Yuda Feng, Yaxiong Xie, Deepak Ganesan, and Jie Xiong. 2021. LTE-Based Pervasive Sensing Across Indoor and Outdoor. In *Proceedings of the 19th ACM Conference on Embedded Networked Sensor Systems*. 138–151. doi:10.1145/3485730.3485943

[10] Yuda Feng, Yaxiong Xie, Deepak Ganesan, and Jie Xiong. 2022. LTE-Based Low-Cost and Low-Power Soil Moisture Sensing. In *Proceedings of the 20th ACM Conference on Embedded Networked Sensor Systems*. 421–434. doi:10.1145/3560905.3568525

[11] Bandara Gajanayake, K. Raja Reddy, Mark W. Shankle, and Ramón A. Arancibia. 2013. Early-season soil moisture deficit reduces sweetpotato storage root initiation and development. *HortScience* 48, 12 (2013), 1457–1462. doi:10.21273/HORTSCI.48.12.1457

[12] Bandara Gajanayake, K. Raja Reddy, Mark W. Shankle, and Ramón A. Arancibia. 2014. Growth, developmental, and physiological responses of two sweetpotato (Ipomoea batatas L. [Lam]) cultivars to early season soil moisture deficit. *Scientia Horticulturae* 168 (2014), 218–228. doi:10.1016/j.scienta.2014.01.018

[13] Martti T. Hallikainen, Fawwaz T. Ulaby, Myron C. Dobson, Mohamed A. El-Rayes, and Lin-Kun Wu. 1985. Microwave dielectric behavior of wet soil—Part I: Empirical models and experimental observations. *IEEE Transactions on Geoscience and Remote Sensing* GE-23, 1 (1985), 25–34. doi:10.1109/TGRS.1985.289497

[14] Kevin A. Handreck and Neil D. Black. 2002. *Growing Media for Ornamental Plants and Turf*. UNSW Press, Sydney.

[15] Dewayne L. Ingram. 2014. *Understanding Soilless Media Test Results and Their Implications on Nursery and Greenhouse Crop Management*. Technical Report HO-112. University of Kentucky Cooperative Extension Service, Lexington, KY. https://publications.mgcafe.uky.edu/sites/publications.ca.uky.edu/files/HO112.pdf

[16] Wenli Jiao, Ju Wang, Yelu He, Xiangdong Xi, and Fuwei Wang. 2024. SoilTAG: Fine-Grained Soil Moisture Sensing Through Chipless Tags. *IEEE Transactions on Mobile Computing* 23, 3 (2024), 2153–2170. doi:10.1109/TMC.2023.3253135

[17] Usman Mahmood Khan and Muhammad Shahzad. 2022. Estimating Soil Moisture Using RF Signals. In *Proceedings of the 28th Annual International Conference on Mobile Computing and Networking*. 242–254. doi:10.1145/3495243.3517025

[18] F. Kizito, C. S. Campbell, G. S. Campbell, D. R. Cobos, B. L. Teare, B. Carter, and J. W. Hopmans. 2008. Frequency, Electrical Conductivity and Temperature Analysis of a Low-Cost Capacitance Soil Moisture Sensor. *Journal of Hydrology* 352, 3–4 (2008), 367–378. doi:10.1016/j.jhydrol.2008.01.021

[19] Anchang Li, Lingxiao Zhu, Wenjun Xu, Liantao Liu, and Guifa Teng. 2022. Recent advances in methods for in situ root phenotyping. *PeerJ* 10 (2022), e13638. doi:10.7717/peerj.13638

[20] Xiuwei Liu, Xuejun Dong, Qingwu Xue, Daniel I. Leskovar, John Jifon, John R. Butnor, and Thomas Marek. 2018. Ground-Penetrating Radar (GPR) Detects Fine Roots of Agricultural Crops in the Field. *Plant and Soil* 423, 1–2 (2018), 517–531. doi:10.1007/s11104-017-3531-3

[21] Federico Lombardi, Bianca Ortuani, Arianna Facchi, and Maurizio Lualdi. 2022. Assessing the Perspectives of Ground Penetrating Radar for Precision Farming. *Remote Sensing* 14, 23 (2022), 6066. doi:10.3390/rs14236066

[22] Siddhesh Mane, Narendra Das, Gurjeet Singh, Michael H. Cosh, and Younsuk Dong. 2024. Advancements in Dielectric Soil Moisture Sensor Calibration: A Comprehensive Review of Methods and Techniques. *Computers and Electronics in Agriculture* 218 (2024), 108686. doi:10.1016/j.compag.2024.108686

[23] NC State University Office of Research and Innovation. 2026. Phytotron Facility. https://research.ncsu.edu/phytotron-facility/. Accessed 2026-03-14.

[24] NC State University Office of Research and Innovation. 2026. Plant Sciences Building Growth Facilities. https://research.ncsu.edu/plant-sciences-building-growth-facilities/. Accessed 2026-03-14.

[25] NC State University Phytotron. 2026. General Information. https://phytotron.ncsu.edu/general-information/. Accessed 2026-03-14.

[26] NC State University Phytotron. 2026. Greenhouses. https://phytotron.ncsu.edu/greenhouses-copy/. Accessed 2026-03-14.

[27] W. Garrett Owen and Roberto G. Lopez. 2015. *Evaluating Container Substrates and Their Components*. Technical Report HO-255-W. Purdue University Extension, West Lafayette, IN. https://www.extension.purdue.edu/extmedia/HO/HO-255-W.pdf

[28] Kalidas Pati, Raja Kaliyappan, Alok Kumar Giri, V. B. S. Chauhan, H. Gowda, R. Arutselvan, and Kuttumu Laxminarayana. 2024. Phenological growth stages of sweet potato (Ipomoea batatas (L.) Lam.) according to the extended BBCH scale. *Annals of Applied Biology* 184, 3 (2024), 382–390. doi:10.1111/aab.12894

[29] Michael Raviv and J. Heinrich Lieth (Eds.). 2008. *Soilless Culture: Theory and Practice*. Elsevier, Amsterdam.

[30] Matthew Reynolds, Owen K. Atkin, Malcolm Bennett, Mark Cooper, Ian C. Dodd, M. John Foulkes, Claus Frohberg, Graeme Hammer, Ian R. Henderson, Bingru Huang, Viktor Korzun, Susan R. McCouch, Carlos D. Messina, Barry J. Pogson, Gustavo A. Slafer, Nicolas L. Taylor, and Peter E. Wittich. 2021. Addressing Research Bottlenecks to Crop Productivity. *Trends in Plant Science* 26, 6 (2021), 607–630. doi:10.1016/j.tplants.2021.03.011

[31] David A. Robinson, Colin S. Campbell, Jan W. Hopmans, Brian K. Hornbuckle, Steven B. Jones, Rosemary Knight, Fred Ogden, John Selker, and Olaf Wendroth. 2008. Soil Moisture Measurement for

Ecological and Hydrological Watershed-Scale Observatories: A Review. *Vadose Zone Journal* 7, 1 (2008), 358–389. doi:10.2136/vzj2007.0143

[32] David A. Robinson, Scott B. Jones, J. M. Wraith, Dani Or, and Sharon P. Friedman. 2003. A review of advances in dielectric and electrical conductivity measurement in soils using time domain reflectometry. *Vadose Zone Journal* 2, 4 (2003), 444–475. doi:10.2136/vzj2003.4440

[33] Mark S. Seyfried and Laura E. Grant. 2007. Temperature Effects on Soil Dielectric Properties Measured at 50 MHz. *Vadose Zone Journal* 6, 4 (2007), 759–765. doi:10.2136/vzj2006.0188

[34] Mika Tei, Fumiyuki Soma, Ettore Barbieri, et al. 2024. Non-destructive real-time monitoring of underground root development with distributed fiber optic sensing. *Plant Methods* 20, 1 (2024), 36. doi:10.1186/s13007-024-01160-z

[35] G. C. Topp, J. L. Davis, and A. P. Annan. 1980. Electromagnetic determination of soil water content: Measurements in coaxial transmission lines. *Water Resources Research* 16, 3 (1980), 574–582. doi:10.1029/WR016i003p00574

[36] Tao Wang, Yang Zhao, Zhibin Huang, Jie Liu, and Yang Zhong. 2026. RTS: Towards Underground Root Tuber Sensing via RF Sensor Networks. *ACM Transactions on Sensor Networks* 22, 3 (2026), 1–31. doi:10.1145/3796515

[37] Tao Wang, Yang Zhao, Jinghua Wang, Zhibin Huang, Jie Liu, and Qiaorong Wei. 2025. See-Through Soil: Underground Root Tuber Sensing with RF Sensor Networks. *IEEE Transactions on Geoscience and Remote Sensing* 63 (2025), 1–15. doi:10.1109/TGRS.2025.3617880

[38] Allah Wasaya, Xiying Zhang, Qin Fang, and Zongzheng Yan. 2018. Root Phenotyping for Drought Tolerance: A Review. *Agronomy* 8, 11 (2018), 241. doi:10.3390/agronomy8110241

[39] Elizabeth Will and James E. Faust. 2010. *Growing Media for Greenhouse Production.* Technical Report PB1618. The University of Tennessee Agricultural Extension Service, Knoxville, TN. https://trace.tennessee.edu/utk_agexcomhort/28

[40] Fusang Zhang, Jie Xiong, Zhaoxin Chang, Junqi Ma, and Daqing Zhang. 2022. Mobi2Sense: Empowering Wireless Sensing with Mobility. In *Proceedings of the 28th Annual International Conference on Mobile Computing and Networking*. 268–281. doi:10.1145/3495243.3560518

[41] Hongjuan Zhao, Na Wang, Hongli Sun, Lin Zhu, Kai Zhang, Yan Zhang, Jun Zhu, Anchang Li, Zhan Bai, Xinyou Liu, Hezhong Dong, Liantao Liu, and Chengcai Li. 2022. RhizoPot platform: A high-throughput in situ root phenotyping platform with integrated hardware and software. *Frontiers in Plant Science* 13 (2022), 1004904. doi:10.3389/fpls.2022.1004904

[42] Peng-Fei Zhao, Yong-Qian Wang, Shi-Xian Yan, Li-Feng Fan, Zi-Yang Wang, Qiao Zhou, Jie-Peng Yao, Qiang Cheng, Zhong-Yi Wang, and Lan Huang. 2019. Electrical Imaging of Plant Root Zone: A Review. *Computers and Electronics in Agriculture* 167 (2019), 105058. doi:10.1016/j.compag.2019.105058

[43] Tianyue Zheng, Zhe Chen, Jun Luo, Lin Ke, Chaoyang Zhao, and Yaowen Yang. 2021. SiWa: See into Walls via Deep UWB Radar. In *Proceedings of the 27th Annual International Conference on Mobile Computing and Networking*. 323–336. doi:10.1145/3447993.3483258

[44] Wolfgang Zierer, David Rüscher, Uwe Sonnewald, and Sophia Sonnewald. 2021. Tuber and Tuberous Root Development. *Annual Review of Plant Biology* 72 (2021), 551–580. doi:10.1146/annurev-arplant-080720-084456